NANO SENTINEL

Wafer Inspection Report

Inspection ID: NS-000001
Timestamp (UTC): 2026-10-12T13:35:07.289113+00:00

Inspection field	Result
Model	MobileNetV2 V2
True class (dataset example)	Edge-Ring
Predicted class	Edge-Loc
Confidence	96.52%
Inspection status	MANUAL_REVIEW
Decision reason	Known label and model prediction disagree in this validation example. This is a diagnos
Defective die pixels	154
Total die pixels	533
Defect ratio	28.89%
Provisional severity	HIGH
Recommendation	Predicted defect class: Edge-Loc. Review the wafer map and Grad-CAM before decidi

Explainability

Grad-CAM visualizes image regions contributing to a model prediction. It does not prove the physical location of a defect.

Grad-CAM image unavailable.

Limitations

Severity thresholds are provisional and have not been validated against manufacturing quality standards.
The known dataset label and predicted class disagree in this example; manual review is required.